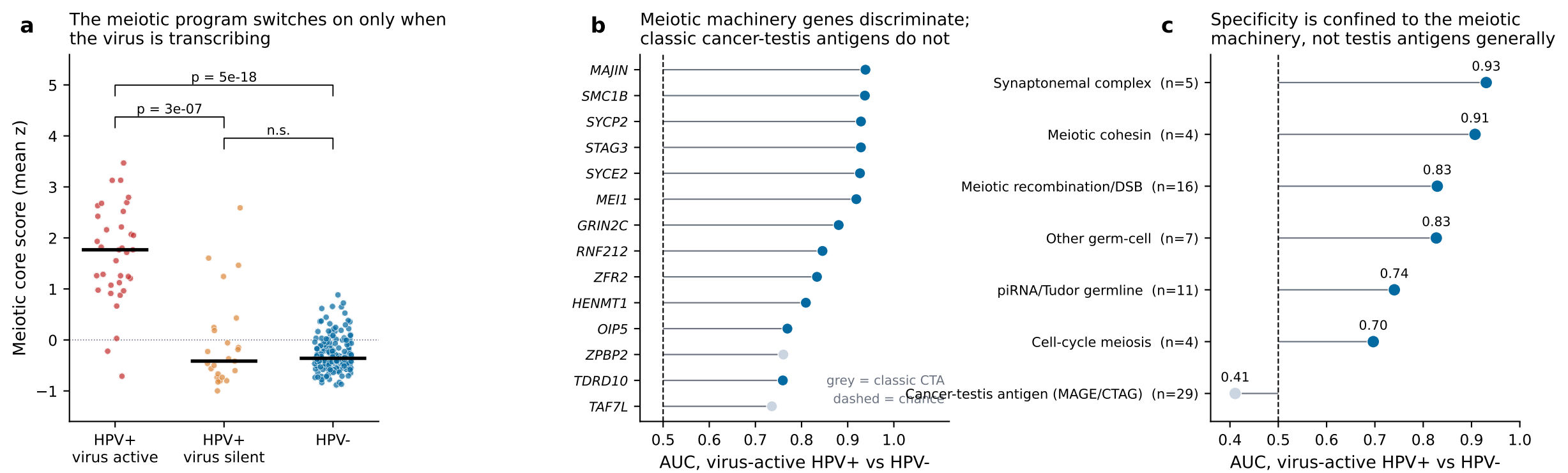
a
The meiotic program switches on only when the virus is transcribing
Meiotic core score (mean z)
p = 5e-18
p = 3e-07
n.s.
HPV+ virus active
HPV+ virus silent
HPV-
b
Meiotic machinery genes discriminate; classic cancer-testis antigens do not
MAJIN
SMC1B
SYCP2
STAG3
SYCE2
MEI1
GRIN2C
RNF212
ZFR2
HENMT1
OIP5
ZPBP2
TDRD10
TAF7L
grey = classic CTA
dashed = chance
AUC, virus-active HPV+ vs HPV-
c
Specificity is confined to the meiotic machinery, not testis antigens generally
Synaptonemal complex (n=5)
Meiotic cohesin (n=4)
Meiotic recombination/DSB (n=16)
Other germ-cell (n=7)
piRNA/Tudor germline (n=11)
Cell-cycle meiosis (n=4)
Cancer-testis antigen (MAGE/CTAG) (n=29)
0.93
0.91
0.83
0.83
0.74
0.70
0.41
AUC, virus-active HPV+ vs HPV-